Pre-requisites Specification

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	3 Marks

Pre-requisites Overview & Description:

Pre-requisites represent the essential developer tools, software runtimes, cloud infrastructure accounts, and API access keys necessary prior to initiating the VoyageAI travel planner project. Fulfilling these pre-requisites ensures seamless environment setup, deterministic AI execution, database connectivity, and zero deployment friction.

1. Software & Local Runtime Requirements:

Tool / Software	Min. Version	Project Role / Purpose	Status
Node.js & npm	v18.0.0+ (LTS)	JavaScript runtime powering React 19, Vite bundler, and Tailwind CSS v4 frontend.	**Verified**
Python & pip	3.11 / 3.12	Asynchronous backend runtime executing FastAPI, Uvicorn, and SQLAlchemy 2.0.	**Verified**
Git	v2.40+	Distributed version control for commit auditing and GitHub repository synchronization.	**Verified**
Cursor IDE / VS Code	Latest	AI-native code authoring environment with multi-terminal support and linting.	**Verified**

2. Cloud Infrastructure & AI API Pre-requisites:

Platform	Configuration / Credentials	Architectural Responsibility	Status
Supabase Cloud	Project ID: yygukmccpwvclabkdrie Port 6543 Transaction Pooler	PostgreSQL 15 database, Row-Level Security (RLS), and JWT authentication.	**Connected**
Groq Cloud LPU	GROQ_API_KEY Model: openai/gpt-oss-120b	Ultra-low latency Language Processing Unit for sub-second JSON itinerary generation.	**Connected**
Vercel Edge	Production Deployment Pipeline SPA Client Rewrites (vercel.json)	Global CDN web hosting delivering ultra-fast React 19 Single Page Application.	**Live**
Render.com	Managed Web Service Python 3.12 / Uvicorn Server	Cloud backend hosting FastAPI REST endpoints with async database connection pooling.	**Live**